ARYA GLOBAL DIGITAL

Complete Software Development Requirements & Business Logic Specification

Project: E-Commerce, Referral Commission and Cashback Management System
Company: Arya Global Digital
Programme: Plan A – Arya Business Partner Programme
Programme: Plan B – Arya Generation Growth Programme
Purpose: Functional and technical specification for AI-assisted software development
Backend: Python – FastAPI
Frontend: Separate frontend application
Database: TiDB Cloud
ORM: SQLAlchemy
Database Migrations: Alembic
Deployment: Render (Frontend and Backend)
Database Hosting: TiDB Cloud
Document Type: Functional and Technical Requirements
Purpose: Developer Implementation Reference

PLAN A – COMPLETE SOFTWARE DEVELOPMENT REQUIREMENT DOCUMENT

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1. PROJECT OBJECTIVE

Develop a complete, secure and production-ready e-commerce website for Arya Global Digital.

The application must support:

- Customer registration and login.
- Product management.
- Product purchasing.
- Unique Billing ID generation.
- Permanent referral codes.
- Direct referral relationships.
- Automatic referral commission calculation.
- Separate commission wallet.
- Cashback management.
- Multiple independent cashback queues.
- Automatic ₹250 cashback contribution processing.
- ₹4,000 cashback completion.
- 48-bulb cashback progress visualization.
- Cashback withdrawal requests.
- Admin approval of cashback withdrawals.
- Customer and admin dashboards.
- Transaction history.
- PDF and Excel reports.

The application must be fully functional, with real backend logic and persistent database operations. A frontend-only prototype is not acceptable.

All financial calculations, Billing ID generation, commission processing, cashback processing and withdrawal operations must be handled securely on the backend.

2. WEBSITE DESIGN AND USER EXPERIENCE

The website should have a premium, modern and professional design inspired by the visual quality of Apple's website.

The goal is to create a clean, elegant and simple user experience.

Do not copy Apple's logo, branding, proprietary assets or exact website layouts. Create an original Arya Global Digital design using similar minimalist design principles.

2.1 Design principles

- Minimalist and premium appearance.
- Clean white and light-coloured backgrounds.
- Elegant typography.
- Large, readable headings.
- Proper spacing between sections.
- Simple navigation.
- Smooth and subtle animations.
- Professional product presentation.
- Rounded cards and clean layouts.
- Consistent colours and components.
- Responsive design for desktop, tablet and mobile.
- Fast page loading.
- Accessible text and controls.

Avoid unnecessary colours, excessive gradients, crowded layouts and distracting animations.

2.2 Website pages

Create the following pages:

1. Home page.
2. Product listing page.
3. Product details page.
4. Customer registration page.
5. Customer login page.
6. Shopping cart and checkout.
7. Payment status page.
8. Customer dashboard.
9. Referral details page.
10. Commission wallet page.
11. Cashback progress page.

12. Withdrawal history page.
13. Admin login page.
14. Admin dashboard.
15. Product management page.
16. Customer management page.
17. Billing ID management page.
18. Commission management page.
19. Cashback queue management page.
20. Cashback withdrawal management page.
21. Reports and exports page.

2.3 Customer experience

The customer should not need to understand the internal commission calculations, FIFO queue implementation or technical cashback processing.

Display only the information needed by the customer in simple language.

The 48-bulb cashback progress indicator must make the cashback process easy to understand.

2.4 Responsive design

The website must work properly on:

- Desktop computers.
- Laptops.
- Tablets.
- Android phones.
- iPhones.

All dashboards, tables, cards and visual progress indicators must adapt to smaller screens.

3. PRODUCT PRICE AND FINANCIAL ALLOCATION

The product selling price is ₹3,999.

The allocation for each product is:

Allocation	**Amount**	**% of Rs. 3,999**
Product import cost	₹1,500	37.5094%
GST reserve	₹400	10.0025%*
Packing and courier	₹150	3.7509%
Customer cashback allocation	₹250	6.2516 %
Gross commission pool	₹750	18.7547%
Gift and achievement fund	₹150	3.7509 %
Office and staff expenses	₹100	2.5006%

Total allocation	**₹3,300**	**82.5206%**
Net company income / balance	**₹699**	**17.4794%**

Important GST rule

The ₹400 GST reserve is calculated using a separate ₹4,000 reference base.

The ₹400 GST reserve is already included in the ₹3,300 total allocation.

Do not deduct the GST reserve again from the ₹750 commission pool.

The financial allocation and tax settings must be configurable if the company receives updated instructions from its accountant or tax adviser.

Commission Deduction Settings

Apply deductions separately to each recipient's gross commission. Rates must be editable in the admin panel.

Deduction / payment	Rate	Calculation base
Operational Cost	4.6667%	Each recipient's gross commission
TDS	2.0000%	Each recipient's gross commission
Total deductions	**6.6667% approx.**	**Each recipient's gross commission**
Net payable	**93.3333% approx.**	**Gross less both deductions**

Formulas

Operational Cost = Gross Commission x
4.6667% TDS = Gross Commission x
2.0000%
Net Payable = Gross Commission minus Operational Cost minus TDS

Use full-precision rates during calculation. Round monetary ledger entries and final payments to two decimal places (paise) using one consistent, documented method. Do not prematurely round intermediate percentage values.

4. CUSTOMER REGISTRATION AND REFERRAL SYSTEM

4.1 Customer registration

1. Customers must be able to register without a referral code.
2. Every registered customer must receive a permanent referral code.
3. Each phone number must be unique.
4. The same phone number must not be used to create multiple customer accounts.
5. If an invalid referral code is entered, display "Invalid Referral".
6. Referral codes must be permanent after registration.
7. Customers must not be able to change their referral person after registration.
8. Self-referral must not be allowed, including through another account or phone number.
9. A customer who has not purchased a Power Bank is inactive.
10. An inactive customer's referral code must not be accepted for a qualifying referral.

4.2 Referral relationship

Plan A supports direct referral commission only.

Example:

- F1 refers F2.
- F2 registers using F1's referral code.
- F2 purchases a qualifying product.
- F1 receives the applicable direct referral commission.

There are no Generation 1–5 commission payouts under Plan A.

The system must automatically maintain referral relationships.

Admin must be able to view the referral relationships.

Admin must not manually change referral relationships under the current business rule.

5. PRODUCT PURCHASE AND UNIQUE BILLING ID GENERATION

This is a critical requirement.

Every successfully paid product must generate its own unique Billing ID.

A Billing ID represents one individual product purchase, not an entire order.

5.1 Billing ID rules

1. Generate a unique Billing ID for every successfully paid product.
2. If a customer purchases one Power Bank, generate one Billing ID.
3. If a customer purchases two Power Banks, generate two separate Billing IDs.
4. If a customer purchases three Power Banks, generate three separate Billing IDs.
5. If an order contains multiple products, generate a separate Billing ID for every product.
6. Billing IDs must never be duplicated.
7. Every Billing ID must be permanently associated with its customer and product purchase.
8. Billing IDs must be stored in the database.
9. Billing IDs must be displayed in the customer's dashboard.
10. Admin must be able to search using a Billing ID.
11. Billing IDs must remain unchanged after generation.
12. Billing ID generation must happen on the backend, not in frontend JavaScript.

5.2 Example

F1 purchases two Power Banks.

The system generates:

- Billing ID 201
- Billing ID 202

F2 purchases three Power Banks.

The system generates:

- Billing ID 203
- Billing ID 204

- Billing ID 205

All five Billing IDs must be separate database records.

5.3 Technical requirements

Use a globally unique Billing ID generation mechanism.

The system must guarantee uniqueness when multiple customers make purchases simultaneously.

Use database constraints to prevent duplicate Billing IDs.

The Billing ID generation process must be idempotent.

If payment succeeds but the application temporarily fails to generate a Billing ID, the backend must be able to recover and complete the missing operation without creating duplicate records.

Do not use hardcoded Billing IDs.

6. PLAN A – REFERRAL COMMISSION CALCULATION

Under Plan A, the eligible direct referrer receives the entire ₹750 gross commission pool for each qualifying product purchased by the referred customer.

6.1 Commission calculation per product

Description	**Amount**
Gross commission	₹750
Operational cost (4.6667%)	₹35
TDS (2.0000%)	₹15
Net commission	**₹700**

Formula:

₹750 - ₹35 - ₹15 = ₹700.

6.2 Commission rules

1. Commission must be calculated separately for every qualifying product.
2. Commission must not be calculated only once per order when the order contains multiple products.
3. Each product-level Billing ID must have its own commission transaction.
4. Commission must be generated after successful payment and Billing ID generation.
5. Commission must be credited to the eligible direct referrer's commission wallet.
6. Commission must be associated with the relevant purchase and Billing ID.
7. Commission generation must not require admin approval.
8. Commission must show gross commission, operational cost, TDS and net commission separately.
9. Each commission transaction must have a unique transaction reference.
10. Duplicate commission transactions must be prevented.

6.3 Example: F1 refers F2

F1 purchases two Power Banks.

F1 receives:

- Billing ID 201
- Billing ID 202

F1 refers F2.

F2 purchases three Power Banks.

F2 receives:

- Billing ID 203
- Billing ID 204
- Billing ID 205

F1 receives commission for all three qualifying products purchased by F2.

F2 product	Billing ID	F1 net commission
Power Bank 1	203	₹700
Power Bank 2	204	₹700
Power Bank 3	205	₹700
Total		**₹2,100**

F1 receives ₹2,100 net commission.

Each ₹700 must appear as a separate wallet transaction.

7. REFERRAL COMMISSION WALLET AND WITHDRAWAL

Commission and cashback must have separate balances and transaction histories.

Commission wallet rules

1. Net commission is credited automatically after successful payment.
2. Customers can view their commission wallet balance.
3. Customers can view individual commission transactions.
4. Customers can request withdrawal of available commission independently.
5. Commission withdrawal does not require completion of 16 cashback contributions.
6. Commission generation does not require admin approval.
7. Commission withdrawal must not depend on cashback cycle status.
8. Every withdrawal request must have a unique reference.
9. The system must prevent duplicate withdrawal requests.
10. Withdrawal history must be available to the customer.

The commission withdrawal process and actual transfer method can be configured separately.

8. CASHBACK SYSTEM

Every eligible product-level Billing ID must be considered separately for cashback processing according to the approved queue assignment and contributor eligibility rules.

Each cashback recipient Billing ID has a target of ₹4,000.

Each eligible contributing Billing ID contributes ₹250.

8.1 Cashback calculation

Description	Amount
Required contributions	16
Contribution per eligible Billing ID	₹250
Total cashback target	₹4,000

Calculation:

16 × ₹250 = ₹4,000.

8.2 Automatic cashback contribution

The contribution must be triggered by a successful product payment and Billing ID generation.

The process must work as follows:

1. Customer completes a successful product payment.
2. Backend verifies the successful payment.
3. System generates a unique Billing ID.
4. System detects the newly generated Billing ID.
5. Cashback engine identifies the next eligible cashback recipient in the applicable independent queue.
6. System automatically records a ₹250 contribution against that recipient's cashback cycle.
7. Contribution count increases by one.
8. Cashback balance increases by ₹250.
9. Cashback progress is updated.
10. The customer dashboard receives the updated progress.

There must not be a separate manual ₹250 payment step after a qualifying successful purchase.

If payment fails, no Billing ID or cashback contribution must be generated.

8.3 Cashback example

Billing ID 201 is the active cashback recipient.

Suppose it receives three eligible contributions.

Contribution	Amount
First contribution	₹250
Second contribution	₹250
Third contribution	₹250
Total	**₹750**

The system must record:

- Recipient Billing ID: 201
- Contributions received: 3/16
- Cashback balance: ₹750

- Remaining contributions: 13
- Remaining amount: ₹3,250
- Cycle status: ACTIVE

After 16 successful contributions:

16 × ₹250 = ₹4,000.

The cashback cycle becomes COMPLETED.

9. MULTIPLE INDEPENDENT CASHBACK QUEUES

This is a mandatory system requirement.

The application must support two or more independent cashback queues.

Each queue must maintain its own sequence, Billing IDs, queue positions, contribution records, cashback balance, contribution count and status.

One queue must never interrupt, reset, modify or incorrectly advance another queue.

9.1 Example: F1 cashback queue

F1 purchases two Power Banks.

F1 receives:

- Billing ID 201
- Billing ID 202

F1 refers F2.

F2 purchases three Power Banks.

F2 receives:

- Billing ID 203
- Billing ID 204
- Billing ID 205

Example queue:

Queue position	Billing ID	Owner
1	201	F1
2	202	F1
3	203	F2
4	204	F2
5	205	F2

Billing ID 201 is the first cashback recipient in this example.

If Billing ID 201 has received three eligible contributions, its progress is 3/16 and its cashback balance is ₹750.

The remaining queue records must remain available for their own processing.

9.2 Example: J1 starts another independent queue

A new customer J1 purchases one Power Bank.

The system generates:

- Billing ID 206

J1 refers J2.

J2 purchases three Power Banks.

The system generates:

- Billing ID 207
- Billing ID 208
- Billing ID 209

A separate cashback queue is created for this independent sequence.

Queue position	Billing ID	Owner
1	206	J1
2	207	J2
3	208	J2
4	209	J2

Billing ID 206 is the first cashback recipient in this example.

9.3 Independent queue behaviour

The application must maintain the F1 and J1 queues independently.

For example:

- A contribution assigned to the F1 queue must update only the F1 queue.
- A contribution assigned to the J1 queue must update only the J1 queue.
- Completing F1's cashback cycle must not reset J1's progress.
- Completing J1's cashback cycle must not change F1's progress.
- Each queue must advance independently after its active cashback cycle is completed.
- New eligible Billing IDs must be assigned to the correct queue according to the approved queue assignment rules.

9.4 Queue implementation

Use FIFO (First In, First Out) logic for recipient ordering.

The first eligible recipient in a queue must be processed first.

Use a persistent database-backed queue.

Do not rely only on an in-memory Python list or queue.

Suggested database fields:

- queue_id
- queue_position
- billing_id
- customer_id

- contribution_count
- cashback_amount
- status
- created_at
- updated_at

Suggested statuses:

- WAITING
- ACTIVE
- COMPLETED

The queue must retain its state after application restarts and redeployments.

Use database transactions and appropriate locking to prevent simultaneous purchases from assigning contributions incorrectly.

10. CUSTOMER DASHBOARD – 48-BULB CASHBACK PROGRESS INDICATOR

This is a mandatory frontend requirement.

The customer-facing dashboard must display cashback progress using a visual indicator containing 48 bulbs for each active cashback cycle.

The customer should not need to understand the internal FIFO queue or the 16-contribution processing logic.

10.1 Bulb calculation

- Every cashback cycle contains 48 bulbs.
- Each successful ₹250 contribution switches ON exactly 3 bulbs.
- Each contribution represents 3 visual bulbs.
- After 16 successful contributions, all 48 bulbs must be ON.
- A completed cycle represents ₹4,000 cashback.

Calculation:

16 contributions × 3 bulbs = 48 bulbs.

10.2 Example: Billing ID 201

Suppose Billing ID 201 receives three eligible contributions.

Description	Value
Contributions received	3/16
Cashback balance	₹750
Bulbs ON	9/48
Bulbs OFF	39/48
Cycle status	ACTIVE

The dashboard must display:

- 9 green ON bulbs.

- 39 OFF bulbs.
- Progress text: 9/48.
- Cashback balance: ₹750.

The frontend must visually represent the actual contribution count received from the backend.

10.3 Visual design

The 48 bulbs should be arranged in a clean and attractive grid.

Requirements:

- ON bulbs must be displayed in green.
- OFF bulbs must be displayed in a neutral grey.
- Bulbs must be equally sized.
- Bulbs must have consistent spacing.
- The design must work on desktop and mobile.
- Progress must update automatically after a valid contribution is recorded.
- Use subtle animations when bulbs switch ON.
- Do not use distracting or excessive animations.

10.4 Sequential cashback cycle display

The customer dashboard must display the current active cashback recipient's 48-bulb progress.

Example:

1. Billing ID 201 is active.
2. The dashboard shows Billing ID 201 and its 48 bulbs.
3. When Billing ID 201 completes all 48 bulbs, its cycle becomes COMPLETED.
4. The dashboard moves to Billing ID 202 and displays its 48-bulb progress.
5. When Billing ID 202 completes, the dashboard moves to Billing ID 203.
6. The same process continues for Billing IDs 204, 205 and subsequent eligible Billing IDs.

When a cycle is completed, the completed cycle must be retained in transaction history.

The customer-facing current progress area should display the next active cycle rather than unnecessarily displaying every completed cycle.

10.5 Backend and frontend separation

- Backend must track the actual 16 successful ₹250 contributions.
- Frontend must convert each contribution into 3 visual ON bulbs.
- The 48-bulb display is only a visual representation.
- It must not replace actual financial contribution records.
- The frontend must receive the correct progress from the backend API.
- Never calculate or credit actual cashback based only on the number of bulbs displayed.
- The backend is the source of truth for contribution count, cashback amount and cycle status.

11. CASHBACK WITHDRAWAL

Cashback withdrawal is different from commission withdrawal.

Rules

1. A cashback cycle requires 16 valid contributions.
2. The completed cashback amount is ₹4,000.
3. The customer/business partner can request withdrawal only after cycle completion.
4. Partial cashback withdrawal is not allowed.
5. The withdrawal request must be for the full ₹4,000.
6. Admin approval is required.
7. Admin can approve or reject the request.
8. If rejected, an explanation must be recorded.
9. If rejected, the ₹4,000 balance must remain visible.
10. The customer must be able to view cashback withdrawal history.
11. Only one active withdrawal request must be allowed for the same completed cycle.
12. Admin must receive a notification when a cashback cycle is completed.
13. Admin must receive a notification when a cashback withdrawal request is submitted.

Maintain permanent records of withdrawal requests and admin decisions.

12. CUSTOMER DASHBOARD

Provide a clean, modern and responsive customer dashboard.

The dashboard must show the following information.

12.1 Customer profile

- Customer name.
- Registered phone number.
- Permanent referral code.
- Account status.

12.2 Purchase history

- Product name.
- Product quantity.
- Order reference.
- Unique Billing IDs.
- Purchase date.
- Payment status.

12.3 Referral information

- Directly referred customers.
- Referral history.
- Eligible referral commission transactions.

12.4 Commission wallet

- Gross commission.
- Operational cost.
- TDS.
- Net commission.
- Available commission balance.
- Commission transaction history.
- Commission withdrawal history.

12.5 Cashback dashboard

- Current active cashback Billing ID.
- 48-bulb progress indicator.
- Number of bulbs switched ON.
- Number of bulbs remaining OFF.
- Cashback amount received.
- Cashback target of ₹4,000.
- Cycle completion status.
- Cashback withdrawal request status.

Customers must be able to view their transaction history.

Customers must not be provided with downloadable commission or cashback statements under the current requirement.

13. ADMIN DASHBOARD

The application must have one administrator account.

Admin must have access to the complete system.

13.1 Admin capabilities

1. View total customers.
2. Search customers by name.
3. Search customers by phone number.
4. Search using Billing ID.
5. View complete product-wise purchase history.
6. View customer referral relationships.
7. View all generated Billing IDs.
8. View all commission transactions.
9. View commission deductions.
10. View all cashback queues.
11. View individual cashback contributions.
12. View each cashback recipient's contribution count.

13. View completed and active cashback cycles.
14. View all 48-bulb progress calculations.
15. View cashback withdrawal requests.
16. Approve or reject cashback withdrawal requests.
17. Record rejection reasons.
18. View commission and cashback transaction histories.
19. Download individual customer reports.
20. Export separate commission and cashback reports in Excel and PDF formats.

13.2 Admin notifications

Admin must receive notifications for:

- Successful cashback cycle completion.
- New cashback withdrawal requests.
- Important transaction failures requiring administrative attention.

Admin must be able to inspect the related customer, Billing ID and transaction reference.

13.3 Admin restrictions

- Admin must not manually change referral relationships under the current rule.
- Admin approval is required for cashback withdrawal only.
- Referral commission must be generated automatically without admin approval.
- Admin must not be able to create duplicate financial transactions through repeated actions.

14. REPORTS AND TRANSACTION HISTORY

Provide separate reports for:

1. Product sales.
2. Billing ID generation.
3. Referral commission.
4. Commission deductions.
5. Commission withdrawals.
6. Cashback contributions.
7. Cashback queues.
8. Cashback cycles.
9. Cashback withdrawals.
10. Customer-wise transaction history.

Reports must support appropriate date, customer, queue and Billing ID filters.

Provide PDF and Excel export options for admin reports.

Every transaction must have a unique reference.

Never silently delete or overwrite financial transaction history.

Maintain an audit trail for admin actions and financial status changes.

15. DATABASE REQUIREMENTS – TIDB CLOUD

Use TiDB Cloud as the persistent production database.

TiDB is MySQL-compatible.

Use a suitable MySQL-compatible SQLAlchemy database driver.

Suggested database entities:

1. users
2. products
3. orders
4. order_items
5. payments
6. billing_ids
7. referral_relationships
8. commission_transactions
9. commission_wallets
10. commission_withdrawals
11. cashback_queues
12. cashback_queue_items
13. cashback_cycles
14. cashback_contributions
15. cashback_withdrawals
16. admin_users
17. notifications
18. audit_logs

The final schema may be normalized and adjusted by the developer while preserving all business rules.

15.1 Database rules

- Use primary keys and foreign keys.
- Enforce globally unique Billing IDs.
- Enforce unique payment references where applicable.
- Store customer ownership for every Billing ID.
- Store the queue associated with each Billing ID.
- Store each ₹250 contribution as an individual transaction.
- Prevent duplicate contributions.
- Store commission and cashback in separate ledgers.
- Use appropriate database indexes.

- Maintain transaction timestamps.
- Store monetary values using DECIMAL or integer paise.
- Do not use binary floating-point values for financial calculations.
- Preserve transaction history.

15.2 Queue persistence

The cashback queues must be database-backed.

Queue state must survive:

- Backend restart.
- Render redeployment.
- Application crash.
- Database reconnection.
- Server restart.

Never use only an in-memory queue as the production source of truth.

16. PAYMENT AND FINANCIAL TRANSACTION SAFETY

The backend must process payment events safely.

Mandatory requirements:

1. Verify payment status through the payment provider's trusted server-side verification mechanism.
2. Do not trust a frontend success message as proof of payment.
3. Generate Billing IDs only for verified successful payments.
4. Prevent duplicate Billing ID creation.
5. Prevent duplicate commission credits.
6. Prevent duplicate cashback contributions.
7. Use idempotency keys or unique transaction references.
8. Use database transactions for related financial operations.
9. Handle concurrent successful purchases safely.
10. Maintain permanent transaction history.

If a payment provider sends the same successful payment notification multiple times, the system must process the payment only once.

If a temporary failure occurs during Billing ID generation, commission creation or cashback contribution processing, the system must safely retry or recover the incomplete operation.

Financial operations must not be partially committed in a way that creates incorrect balances.

17. REFUND AND CANCELLATION RULE

Under the current Plan A business instruction:

- Product purchases are treated as final after successful payment.
- Customer cancellation is not allowed.
- Customer refunds are not allowed.

- Commission reversal due to customer cancellation or refund is not required under this Plan A specification.

Do not implement a customer-facing cancellation or refund workflow unless the company provides a revised written requirement.

18. SECURITY REQUIREMENTS

Implement appropriate application security.

1. Secure password hashing.
2. Authenticated customer access.
3. Authenticated admin access.
4. Role-based authorization.
5. Secure API endpoints.
6. Input validation.
7. SQL injection protection through safe ORM/database operations.
8. Proper CORS configuration.
9. Secure handling of payment credentials.
10. Environment-variable-based secret management.
11. HTTPS in production.
12. Protection against unauthorized wallet modifications.
13. Server-side validation of all financial transactions.
14. Rate limiting on sensitive authentication endpoints where appropriate.
15. Secure session or token management.

Customers must never be able to modify their commission balance, cashback balance, Billing IDs or financial transaction records directly.

19. TECHNOLOGY STACK

Use the following technology stack unless a change is approved.

Component	**Technology**
Backend language	Python
Backend framework	FastAPI
Database	TiDB Cloud
ORM	SQLAlchemy
Database driver	PyMySQL or compatible MySQL driver
Database migrations	Alembic
Frontend	Separate frontend application
API communication	REST API
Request validation	Pydantic

Authentication	Secure token-based authentication
Version control	Git and GitHub
Backend hosting	Render
Frontend hosting	Render
Database hosting	TiDB Cloud

The backend must expose properly documented REST APIs.

Keep the code modular.

Separate authentication, products, orders, payments, Billing IDs, commissions, cashback queues, withdrawals, reports and admin operations into appropriate modules.

20. DEPLOYMENT REQUIREMENTS – RENDER AND TIDB CLOUD

The application will be deployed using Render and TiDB Cloud.

20.1 Backend deployment

Deploy the FastAPI backend as a Render Web Service.

Requirements:

- Configure the correct production start command.
- Configure the production port using the Render-provided PORT environment variable.
- Configure production environment variables in Render.
- Connect the backend securely to TiDB Cloud.
- Enable required database TLS/SSL certificate verification.
- Configure production CORS origins.
- Ensure the backend starts correctly after deployment.
- Do not store production credentials inside source code.
- Verify that API endpoints work after deployment.

20.2 Frontend deployment

Deploy the frontend separately on Render.

Requirements:

- Configure the production backend API URL.
- Configure frontend environment variables correctly.
- Ensure customer registration and login work.
- Ensure product purchasing works.
- Ensure customer dashboards work.
- Ensure admin dashboards work.
- Ensure all frontend API calls communicate with the deployed backend.
- Verify responsive layouts after deployment.

20.3 TiDB Cloud database

Use TiDB Cloud for persistent production data.

Requirements:

- Configure the production database name.
- Configure secure database credentials.
- Configure SSL/TLS connection settings.
- Run Alembic migrations in a controlled deployment process.
- Verify that all expected tables exist.
- Test database connectivity.
- Verify that records remain available after backend redeployment.
- Never expose database credentials in frontend code or GitHub.

20.4 Environment variables

Use environment variables for configuration, including:

- DATABASE_URL or individual database connection settings.
- Database host.
- Database port.
- Database name.
- Database username.
- Database password.
- SSL certificate path or supported SSL configuration.
- Authentication secret key.
- CORS origins.
- Payment gateway credentials.

Use a .env.example file containing placeholder values only.

Never commit real .env files or production secrets to GitHub.

21. TESTING AND ACCEPTANCE REQUIREMENTS

The system must be tested before production deployment.

21.1 Billing ID tests

- One product purchase generates one Billing ID.
- Two product purchases generate two separate Billing IDs.
- Three product purchases generate three separate Billing IDs.
- Simultaneous purchases do not generate duplicate Billing IDs.
- A failed payment does not generate a Billing ID.

21.2 Referral commission tests

- A qualifying direct referral generates ₹750 gross commission.
- Operational cost is ₹35.

- TDS is ₹15.
- Net commission is ₹700.
- Three qualifying products generate three separate ₹700 transactions.
- Commission is credited automatically.
- Commission withdrawal is independent of the 16-contribution cashback requirement.
- Duplicate payment notifications do not create duplicate commission transactions.

21.3 Cashback tests

- Each eligible contribution is ₹250.
- Sixteen successful contributions total ₹4,000.
- Three contributions produce ₹750 and progress of 3/16.
- Three contributions switch ON exactly 9 of 48 bulbs.
- Sixteen contributions switch ON all 48 bulbs.
- The cashback cycle completes only after 16 valid contributions.
- The recipient can request ₹4,000 after completion.
- Partial withdrawal is not allowed.
- Cashback withdrawal requires admin approval.
- Rejected withdrawal requests retain the cashback balance.

21.4 Independent queue tests

- F1 and J1 queues must operate independently.
- A contribution in one queue must not update another queue.
- Completing one queue's cycle must not reset another queue.
- Each queue must maintain its own position and contribution count.
- Queue data must remain available after application restart and redeployment.
- Concurrent purchases must not assign contributions incorrectly.
- Duplicate contribution processing must be prevented.

21.5 Dashboard tests

- The current active Billing ID must be displayed.
- The dashboard must show 48 bulbs.
- Each valid contribution must switch ON 3 bulbs.
- The displayed cashback amount must match backend records.
- When a cycle completes, the next active Billing ID must appear.
- Completed cycle history must remain available.
- The dashboard must work on desktop and mobile.

21.6 Admin and customer tests

- Customer can view their Billing IDs.

- Customer can view commission transactions.
- Customer can view cashback progress.
- Admin can search by customer name, phone number and Billing ID.
- Admin can view all queues and contributions.
- Admin can approve or reject cashback withdrawals.
- Admin reports must export correctly.

21.7 Production tests

- Frontend loads successfully.
- Backend API works successfully.
- Backend connects securely to TiDB Cloud.
- Database migrations complete successfully.
- Login and authentication work.
- Customer purchase flow works.
- Financial records are stored correctly.
- No production secrets are exposed.

22. DEVELOPMENT INSTRUCTIONS FOR CHATGPT AND AI CODING TOOLS

This document will be added to the ChatGPT project used during software development.

The AI coding assistant must follow these instructions.

1. Read and understand this complete Plan A document before making code changes.
2. Inspect the existing repository and understand its current structure.
3. Do not rewrite working code unnecessarily.
4. Do not invent new commission, referral or cashback business rules.
5. Do not implement Plan B as part of this development.
6. Implement backend business logic before relying on frontend calculations.
7. Use database-backed queues.
8. Generate unique Billing IDs on the backend.
9. Keep commission and cashback transactions separate.
10. Ensure financial operations are idempotent and transaction-safe.
11. Write unit tests for important financial and queue operations.
12. Test multiple customers and multiple product purchases.
13. Test multiple independent cashback queues.
14. Test the 48-bulb progress calculation.
15. Check database migrations before deployment.
16. Review security and environment configuration.
17. Provide a clear explanation of every major implementation step.

18. Do not claim a feature is complete until it has been implemented and tested.
19. Do not replace real database operations with temporary in-memory data for production functionality.
20. Do not use hardcoded customer IDs, Billing IDs or financial balances.
21. Preserve existing customer and transaction data during code changes and deployments.
22. Follow the premium, minimalist and responsive website design requirements.
23. Do not expose internal queue complexity to customers.
24. Ask for clarification whenever a business rule is unclear or contradictory.

Development should proceed in small, verifiable stages.

After each major stage, run the relevant tests and report:

- What was implemented.
- Which files were changed.
- Which tests were executed.
- Which tests passed.
- Which issues remain.

23. FINAL DELIVERY EXPECTATIONS

The final application must include:

- Premium, responsive e-commerce website.
- Customer registration and login.
- Admin login and dashboard.
- Product management.
- Product purchasing.
- Successful payment verification.
- Unique Billing ID generation.
- Direct referral management.
- Automatic ₹700 net commission credit.
- Independent commission wallet.
- Commission withdrawal functionality.
- Independent cashback balances and withdrawal process.
- FIFO cashback queue processing.
- Multiple independent cashback queues.
- Automatic ₹250 contribution processing.
- ₹4,000 cashback cycle completion.
- 48-bulb cashback progress indicator.
- Automatic movement to the next active cashback cycle.
- Admin cashback withdrawal approval.

- Customer transaction history.
- Admin reports and exports.
- Secure TiDB Cloud database integration.
- Working Render frontend deployment.
- Working Render backend deployment.
- Database migration instructions.
- Testing evidence.
- Basic project and deployment documentation.

The application must be fully functional from customer registration through purchase, Billing ID generation, commission credit, cashback contribution, cashback cycle completion and withdrawal request.

A frontend-only prototype or dashboard without working backend and database functionality is not acceptable.

24. TAX AND FINANCIAL COMPLIANCE

The commission rates, operational cost, TDS and GST reserve in this specification are the requested business calculations.

They are not confirmation that the stated tax rates or treatments are legally applicable.

Before production launch, the company must obtain written confirmation from its accountant or tax adviser.

The software must support appropriate configuration changes if the approved tax treatment changes.

25. IMPORTANT BUSINESS RULES REQUIRING FINAL CONFIRMATION

The following items must be confirmed before implementing the corresponding cashback logic.

25.1 Same-customer contribution restriction

Example:

- F2 purchases three Power Banks.
- Billing IDs 203, 204 and 205 are generated.
- The current example expects all three purchases to contribute ₹250 each to F1's Billing ID 201 cashback cycle.

However, an earlier clarification stated that multiple Billing IDs owned by the same customer must not contribute to the same cashback cycle.

These instructions conflict.

The company must confirm whether Billing IDs 203, 204 and 205 can all contribute separately to Billing ID 201.

Do not silently choose one interpretation.

25.2 Independent queue assignment

The examples show F1 and J1 having separate cashback queues.

The company must define the exact rule for deciding which new Billing ID is assigned to which independent queue, and how contributors are selected for each queue.

Do not assume that every new Billing ID can contribute to every queue.

25.3 Commission wallet ownership

The commission must be credited to the direct referrer's account and associated with the relevant Billing ID.

The developer must preserve the product-level commission history and the customer's account-level available commission balance.

The system must not duplicate a commission when a customer owns multiple Billing IDs.

END OF PLAN A – COMPLETE SOFTWARE DEVELOPMENT REQUIREMENT DOCUMENT

PLAN B – ARYA GENERATION GROWTH PROGRAMME

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1. PROJECT OVERVIEW

Arya Global Digital requires an e-commerce website with two independent business programmes:

1. Plan A – Arya Business Partner Programme
2. Plan B – Arya Generation Growth Programme

This document describes the complete requirements for Plan B.

Plan B must maintain its own:

- Customer registration and login.
- Distributor referral codes.
- Referral relationships.
- Orders and Billing IDs.
- Commission calculations.
- Wallet transactions.
- Withdrawal records.
- Distributor dashboard.
- Admin management and reporting.

Plan A and Plan B must operate as separate programmes. Their commission calculations, referral records, wallets and dashboards must not be mixed.

The website should have a premium, professional and Apple-inspired appearance, with clean typography, elegant spacing, smooth transitions and responsive layouts.

2. PROGRAMME SELECTION DURING REGISTRATION

2.1 Programme Selection

When a new user creates an account, the registration page must display two programme options:

Option 1: Plan A – Arya Business Partner Programme

Option 2: Plan B – Arya Generation Growth Programme

The user must select the programme before completing registration.

The website must clearly explain the selected programme before the user continues.

2.2 Separate Accounts

Plan A and Plan B must maintain separate accounts.

Rules:

1. A user cannot use the same mobile number to register in both programmes.
2. A person may join both programmes using different mobile numbers.
3. Each programme must have independent account information.
4. Each programme must have its own login option.
5. Each programme must have its own dashboard.

6. Commission balances and referral records must remain separate.

2.3 Permanent Programme Selection

Once an account is registered under Plan B:

- The account permanently belongs to Plan B.
- The user cannot transfer the account to Plan A.
- The user cannot transfer referral history.
- The user cannot transfer commission balances.
- Admin cannot change the selected programme.

If a user selects the wrong programme, they may create a new account using a different mobile number.

The new account must be independent. Previous account details, referral relationships and wallet balances must not be transferred.

3. USER REGISTRATION AND LOGIN

3.1 Registration Information

The registration form should collect the required customer information.

Suggested fields:

- Full name
- Mobile number
- Email address
- Password
- Confirm password
- Required address information

The developer should implement appropriate validation for all required fields.

The mobile number must be unique within the Plan B programme.

3.2 Referral Code During Registration

A referral code is not required during account registration.

The referral code will be entered during the ordering process.

3.3 Login

Plan B users must have a separate login option.

After successful login, the user should be redirected to the Plan B distributor dashboard.

3.4 Password Security

- Passwords must never be stored in plain text.
- Use a secure password hashing algorithm.
- Validate login credentials on the backend.
- Protect authenticated routes.
- Implement role-based access control.

4. REFERRAL CODE MANAGEMENT

4.1 Automatic Referral Code Generation

Whenever a Plan B account is successfully created, the system must automatically generate a unique referral code.

The referral code must:

- Be unique.
- Be permanently associated with the account.
- Appear in the user's Profile section.
- Be available for copying.
- Remain fixed throughout the account's lifetime.

Users cannot manually edit or change their referral codes.

Admin cannot manually change referral relationships or replace a customer's referring distributor.

4.2 Referral Code Usage

A Plan B distributor can refer people who choose Plan A during registration.

A Plan A customer can refer other Plan A customers.

A Plan A customer cannot refer someone to Plan B.

Important business rule: Plan B customers must not be able to create Plan B-to-Plan B referrals. Plan B referral codes can be used to refer customers who join Plan A.

4.3 Referral Code Validation During Checkout

During checkout, the customer must enter the referral code.

The system must validate the code through the backend.

If the referral code is invalid, display:

"Wrong Referral ID"

The invalid code must not be accepted.

If the referral code is valid:

1. Validate the code.
2. Retrieve the referring distributor's name.
3. Display the distributor's name.
4. Allow the customer to verify the referral information.
5. Save the referral relationship when the order is successfully placed.

4.4 Permanent Referral Relationship

Once an order is successfully placed:

- The referral relationship becomes permanent.
- Customers cannot change the referral ID.
- Admin cannot change the referral relationship.
- Admin cannot move a customer to another referral hierarchy.

The system must maintain permanent referral records.

5. PLAN B COMMISSION STRUCTURE

5.1 Commission Components

Plan B has two commission components:

1. Direct Referral Commission – ₹500
2. Generation Commission – ₹50

The commission rules are different.

Commission type	Amount	Credit rule
Direct Referral	₹500	Once per referred customer for their lifetime
Generation Commission	₹50	For every eligible Billing ID

The ₹500 commission must not be credited again when the same referred customer makes future purchases.

The ₹50 commission continues for every eligible Billing ID without a fixed lifetime limit.

5.2 Direct Referral Commission – ₹500

The ₹500 Direct Referral commission is credited only once per referred customer.

Example:

1. Distributor 1 refers Customer 1.
2. Customer 1 successfully places their first order.
3. The system generates the Billing ID or Billing IDs.
4. The system checks whether the ₹500 commission has already been credited for Customer 1.
5. If it has not been credited, ₹500 is credited automatically to Distributor 1's wallet.

Future purchases by Customer 1 must not generate another ₹500 Direct Referral commission.

The system must maintain a permanent record of the first Direct Referral commission.

5.3 Generation Commission – ₹50

The ₹50 Generation Commission is credited for every eligible Billing ID.

Example:

Distributor 1 refers Customer 1.

Customer 1 purchases three Power Banks.

The system generates three Billing IDs.

Billing ID	Generation commission
Billing ID 1	₹50
Billing ID 2	₹50
Billing ID 3	₹50
Total	**₹150**

Distributor 1 receives ₹150 Generation Commission.

The commission must be credited automatically after successful order placement and Billing ID generation.

5.4 Combined Commission Example

Distributor 1 refers Customer 1.

Customer 1 purchases three Power Banks in the first order.

Commission component	Calculation	Amount
Direct Referral	One time	₹500
Generation Commission	3 × ₹50	₹150
Total gross commission		**₹650**

Distributor 1 receives ₹650 gross commission.

5.5 Future Purchase Example

Customer 1 purchases another two Power Banks later.

Two new Billing IDs are generated.

Commission component	Calculation	Amount
Direct Referral	Already credited	₹0
Generation Commission	2 × ₹50	₹100
Total gross commission		**₹100**

Distributor 1 receives another ₹100.

The system must never credit the ₹500 Direct Referral commission again for Customer 1.

5.6 Plan B Refers a Plan A Customer

A Plan B distributor can refer a person who selects Plan A during registration.

Example:

1. Distributor 1 belongs to Plan B.
2. Distributor 1 refers Customer 1.
3. Customer 1 selects Plan A during registration.
4. Customer 1 purchases three Power Banks.
5. Three Billing IDs are generated.

Distributor 1 receives:

- ₹500 Direct Referral commission once.
- ₹50 × 3 = ₹150 Generation Commission.

Total Plan B commission for Distributor 1: ₹650 gross.

Customer 1 participates in Plan A and receives Plan A benefits according to the separate Plan A rules.

The same Billing IDs must not receive both Plan A and Plan B commission calculations.

5.7 Commission Calculation Separation

Plan A and Plan B commission calculations must remain separate.

- Plan A referral code: Apply Plan A rules.
- Plan B referral code: Apply the applicable Plan B referral rules.

- Never calculate both plans' commissions for the same Billing ID.

Important developer instruction: The ₹50 Generation Commission must be implemented according to the confirmed direct-referral Billing ID rule above. Do not invent additional ₹50 payments to multiple upstream distributors or assume that every level of a referral tree receives ₹50. Any additional generation-level allocation requires a separate approved business rule.

6. AUTOMATIC COMMISSION CREDITING

Both commission components must be credited automatically.

6.1 ₹500 Direct Referral

The system must:

1. Identify the referring distributor.
2. Identify the referred customer.
3. Check whether the customer has completed a previous successful purchase.
4. Check whether the ₹500 commission has already been credited.
5. Credit ₹500 only if the commission is eligible and has not previously been credited.
6. Save the wallet transaction.

6.2 ₹50 Generation Commission

For each eligible Billing ID:

1. Verify successful order placement.
2. Verify that the Billing ID is unique.
3. Identify the eligible referring distributor.
4. Calculate ₹50.
5. Credit ₹50 to the distributor's wallet.
6. Save the wallet transaction.

6.3 Commission Records

Every commission transaction must have:

- Transaction ID
- Distributor ID
- Customer ID
- Order ID
- Billing ID
- Commission type
- Commission amount
- Credit status
- Credit date and time

The system must maintain permanent commission records.

7. PRODUCT MANAGEMENT

7.1 Product Management Access

Only the admin can add Power Bank products.

Business partners and customers cannot add products.

7.2 Fixed Product Information

The product selling price is fixed at:

₹3,999

The price must not be editable by admin.

The following product information is fixed once configured:

- Product name
- Product description
- Product image
- Selling price

Admin must not be able to modify these fixed product details.

7.3 Stock Display

Customers should see only the stock availability status.

If the product is available:

In Stock

The actual stock quantity must not be displayed to customers.

If the product is unavailable:

Out of Stock

The ordering option must be disabled when the product is out of stock.

The system must not accept an order when stock is unavailable.

The internal stock update and replenishment procedure must be confirmed before implementation because the product stock quantity is not editable under the current rules.

8. ORDER MANAGEMENT

8.1 Ordering Process

The customer ordering process should follow:

1. Customer logs in.
2. Customer selects the Power Bank.
3. Customer checks product information.
4. Customer selects the required quantity.
5. Customer enters the applicable referral code.
6. System validates the referral code.
7. System displays the referring distributor's name.
8. Customer verifies the referral information.
9. Customer reviews the order.
10. Customer sees the no-cancellation, no-return and no-replacement notice.

11. Customer presses Place Order.
12. The current development version treats the order as successfully paid.
13. The system creates separate Billing IDs for each Power Bank.
14. The system calculates eligible commissions.
15. The system credits the applicable commissions automatically.
16. The system displays the order confirmation.

8.2 Current Payment Behaviour

No real payment gateway will be integrated in the current development phase.

For the initial client demonstration:

- Customer clicks Place Order.
- The system treats the order as successfully paid.
- The system creates the order.
- The system generates Billing IDs.
- The system calculates commissions.
- The system updates the wallet.

This is a simulated payment success flow, not an actual financial payment.

Actual payment gateway integration will be decided later.

8.3 Billing ID Generation

Every successfully purchased Power Bank must receive its own unique Billing ID.

Example:

Customer purchases five Power Banks.

Product	Billing ID
Power Bank 1	Unique Billing ID 1
Power Bank 2	Unique Billing ID 2
Power Bank 3	Unique Billing ID 3
Power Bank 4	Unique Billing ID 4
Power Bank 5	Unique Billing ID 5

Billing IDs must be generated by the backend.

The frontend must never generate or decide Billing IDs.

8.4 Duplicate Order Protection

The backend must prevent duplicate processing.

If the same order request is submitted multiple times, the system must not create duplicate Billing IDs or duplicate commission transactions.

Use:

- Unique order references.

- Unique Billing ID constraints.
- Database transactions.
- Backend idempotency checks.
- Duplicate commission protection.

8.5 Order Cancellation and Returns

Under the current business rules:

- Customers cannot cancel orders after successful payment.
- Customers cannot return purchased Power Banks.
- Customers cannot request replacements.

Display this message before the customer places the order:

"Orders cannot be cancelled, returned or replaced after successful payment."

No customer cancellation or commission reversal workflow is required under the current rules.

9. DISTRIBUTOR WALLET

9.1 Wallet Overview

Every Plan B distributor must have an independent wallet.

The wallet should display:

- Total commission earned.
- Direct Referral commission.
- Generation commission.
- Total withdrawn.
- Available wallet balance.
- Withdrawal history.

All wallet amounts must be maintained using decimal-safe financial calculations.

9.2 Wallet Credit

The system automatically credits:

- ₹500 Direct Referral commission, once per referred customer.
- ₹50 Generation commission, for every eligible Billing ID.

Admin cannot manually add money to a distributor's wallet.

Admin cannot manually deduct money from a distributor's wallet.

Only valid commission credits and withdrawal debits can change the wallet balance.

10. WITHDRAWAL RULES

10.1 First Withdrawal

The minimum balance required for the first withdrawal is:

₹750

If the available wallet balance is below ₹750, the distributor cannot make the first withdrawal.

Example:

Available balance	First withdrawal eligibility
₹500	Not eligible
₹700	Not eligible
₹749	Not eligible
₹750	Eligible
₹1,200	Eligible

10.2 Second Withdrawal Onwards

After the first withdrawal is successfully completed, there is no minimum withdrawal limit.

The distributor can withdraw any available balance.

10.3 Full Balance Withdrawal Only

Partial withdrawals are not allowed.

The distributor must withdraw the entire available wallet balance.

Example:

Description	Amount
Available wallet balance	₹1,200
Withdrawal amount	₹1,200
Remaining balance	₹0

10.4 Automatic Withdrawal

For an eligible active distributor:

- No admin approval is required.
- Withdrawal is processed automatically.
- The wallet balance is updated.
- A transaction record is created.

10.5 Withdrawal Confirmation

After successful withdrawal, the dashboard must display:

- Withdrawal success message.
- Withdrawal amount.
- Transaction ID.
- Date and time.
- Updated wallet balance.

The distributor must be able to view previous withdrawal receipts in their dashboard.

11. BLOCKED DISTRIBUTOR RULES

Admin must have the ability to block or deactivate Plan B distributors.

11.1 Blocked Distributor Access

A blocked distributor:

- Can log in.
- Can view previous commission records.
- Can view previous withdrawal records.
- Can view wallet balance.
- Cannot make new referrals.
- Cannot earn new commissions.

11.2 Blocked Distributor Withdrawal

A blocked distributor:

- Cannot withdraw directly.
- Can submit a withdrawal request.
- Must wait for admin review.

Admin can approve or reject the request.

If approved:

- Process the withdrawal.
- Update the wallet balance.
- Record the transaction.

If rejected:

- Keep the money in the wallet.
- Save the rejection status.
- Preserve the withdrawal request history.

Admin must receive an automatic notification whenever a blocked distributor submits a withdrawal request.

11.3 Reactivation

Admin can reactivate a blocked or deactivated account when permitted.

The system must maintain the account status history.

12. WALLET FREEZE AND UNFREEZE

Wallet freezing is separate from account blocking.

Admin can freeze a distributor's wallet without blocking the entire account.

12.1 Wallet Freeze

When a wallet is frozen:

- Distributor can log in.
- Distributor can view wallet balance.
- Distributor can view commission history.
- New eligible commissions can still be recorded.

- Distributor cannot withdraw money.
- Admin must enter a reason for freezing.

12.2 Distributor Visibility

The freeze reason must be visible to the distributor.

The dashboard should display:

- Wallet status: Frozen.
- Freeze reason.
- Freeze date.
- Freeze time.

12.3 Wallet Unfreeze

Admin can unfreeze the wallet.

After unfreezing:

- Wallet status changes to Active.
- Distributor receives an automatic notification.
- Eligible withdrawals become available again.

12.4 Freeze History

Maintain complete freeze/unfreeze history:

- Distributor ID
- Admin ID
- Action
- Reason
- Freeze date and time
- Unfreeze date and time
- Previous status
- Current status

Both admin and distributor should be able to view the relevant history.

13. DISTRIBUTOR DASHBOARD

The Plan B dashboard must provide all important account, commission, wallet and referral information.

13.1 Dashboard Summary

Display:

1. Total commission earned.
2. Direct Referral commission.
3. Generation commission.
4. Total withdrawn.
5. Available wallet balance.

6. Wallet status.
7. Account status.

13.2 Profile Section

Display:

- Full name
- Mobile number
- Email address
- Registration date
- Programme name
- Unique referral code

The referral code must be fixed and cannot be edited.

13.3 Referred Customer List

The distributor must be able to see individual customers referred by them.

Display:

- Customer name
- Mobile number
- Registration date
- Purchase history
- Number of Power Banks purchased
- Billing IDs
- ₹500 Direct Referral commission status
- ₹50 Generation commission transactions

Do not display a total referred-customer count or a combined total number of Power Banks purchased by referred customers on the distributor dashboard.

13.4 Billing ID History

Display:

- Billing ID
- Customer name
- Order ID
- Product name
- Purchase date
- Commission credited
- Commission status

13.5 Commission History

Every commission must be displayed separately.

For ₹500 Direct Referral:

- Commission type
- Customer name
- Transaction ID
- Credit date
- Amount
- Status

For ₹50 Generation Commission:

- Commission type
- Customer name
- Billing ID
- Transaction ID
- Credit date
- Amount
- Status

13.6 Withdrawal History

Display:

- Withdrawal ID
- Amount
- Date and time
- Transaction ID
- Status
- Receipt

13.7 Excel Commission Statement

Distributors must be able to download their commission statement in Excel (.xlsx) format.

Do not provide a PDF download option for the commission statement.

Suggested columns:

- Transaction date
- Customer name
- Billing ID
- Order ID
- Commission type
- Commission amount
- Transaction ID
- Withdrawal amount
- Wallet balance

14. ADMIN DASHBOARD

Admin must have one central dashboard for managing Plan B.

14.1 Distributor Management

Admin can:

- View all Plan B distributors.
- Search by name.
- Search by mobile number.
- Open individual distributor profiles.
- Block distributors.
- Deactivate distributors.
- Reactivate accounts.
- Freeze wallets.
- Unfreeze wallets.

14.2 Individual Distributor Profile

When admin searches for a person and opens their profile, display all relevant details.

Personal details:

- Name
- Mobile number
- Email address
- Registration date
- Account status

Referral details:

- Referral code
- Referring distributor
- Referral hierarchy

Purchase details:

- Order IDs
- Billing IDs
- Product name
- Quantity
- Purchase date
- Payment status

Commission details:

- ₹500 Direct Referral history
- ₹50 Generation Commission history

- Commission transaction IDs
- Credit dates
- Commission status

Wallet details:

- Total commission earned
- Total withdrawn
- Available balance
- Wallet status

Withdrawal details:

- Withdrawal amount
- Transaction ID
- Date and time
- Status
- Request history

Account and wallet management:

- Active/blocked/deactivated status
- Wallet freeze/unfreeze status
- Freeze reason
- Freeze/unfreeze history

Admin must be able to download the individual distributor's complete details in Excel format.

14.3 Admin Search

Admin must be able to search using:

- Person's name
- Mobile number
- Billing ID
- Order ID

14.4 Referral Hierarchy

Admin must be able to view the referral relationships.

Example:

Distributor 1 → Customer 1 → Customer 2 → Customer 3

Display who referred whom and the associated referral relationships.

Admin can inspect the hierarchy but cannot manually change referral relationships.

14.5 Blocked and Deactivated Distributors

Provide separate sections for:

- Active distributors

- Blocked distributors
- Deactivated distributors

Admin can search and inspect individual profiles in each section.

14.6 Admin Dashboard Statistics

Display:

- Total Plan B distributors.
- Total active distributors.
- Total blocked distributors.
- Total deactivated distributors.
- Total successful orders.
- Total Power Banks purchased.
- Total Billing IDs generated.
- Total successful payments.
- Total Plan B sales revenue.
- Daily sales revenue.
- Weekly sales revenue.
- Monthly sales revenue.
- Overall commission payout summary.
- Total outstanding wallet balance.

Do not display the combined lifetime withdrawal amount of all distributors.

Do not display separate daily, weekly and monthly commission breakdowns for ₹500 and ₹50.

14.7 Commission Payout Summary

Admin can view:

- Total ₹500 Direct Referral commissions credited.
- Total ₹50 Generation commissions credited.
- Combined total commission credited.
- Eligible pending commissions, if any.

14.8 Billing ID Commission Inspection

Admin can search for a Billing ID and inspect:

- Billing ID
- Customer name
- Order ID
- Referring distributor
- ₹500 Direct Referral commission status, if applicable
- ₹50 Generation Commission recipient

- Commission amount
- Transaction ID
- Credit date and time

Admin must be able to inspect the complete commission calculation for the Billing ID.

15. ADMIN REPORTS

All requested reports must be available in Excel (.xlsx) format.

15.1 Complete Plan B Purchase Report

Include:

- Customer name
- Customer mobile number
- Distributor name
- Order ID
- Billing ID
- Product name
- Quantity
- Purchase date and time
- Payment status
- Commission details

15.2 Commission Report

Include:

- Distributor name
- Customer name
- Billing ID
- Order ID
- Commission type
- Commission amount
- Transaction ID
- Credit date and time
- Commission status

15.3 Individual Distributor Report

Include all personal, referral, order, Billing ID, commission, wallet and withdrawal details.

15.4 Search and Filter

Admin can filter records using:

- Customer name
- Distributor name

- Mobile number
- Billing ID
- Order ID
- Date range

16. NOTIFICATION REQUIREMENTS

The following notification rules must be implemented.

16.1 Notifications Not Required

Do not send:

- Customer SMS notifications.
- Customer email notifications.
- Commission credit notifications.
- ₹500 commission notifications.
- ₹50 commission notifications.
- Admin notifications for failed payments.
- Admin notifications for successful orders.
- Admin notifications for new Plan B registrations.

16.2 Notifications Required

Implement:

1. Distributor notification when an order is delivered.
2. Admin notification when a blocked distributor submits a withdrawal request.
3. Distributor notification when a frozen wallet is unfrozen.
4. Distributor withdrawal confirmation on successful withdrawal.

Notifications should be visible inside the website dashboard where applicable.

17. WEBSITE PAGES

The website must have the following pages:

17.1 Home Page

Display:

- Arya Global Digital branding.
- Product information.
- Power Bank details.
- Programme information.
- Login and registration options.
- Clear navigation.

17.2 Programme Information

Display the relevant programme information based on the user's selection.

The user should be able to understand the difference between:

- Plan A – Arya Business Partner Programme.
- Plan B – Arya Generation Growth Programme.

17.3 Contact Us

Provide company contact information and an appropriate contact form.

WhatsApp support is not required.

17.4 About Us

Provide information about Arya Global Digital and its business.

17.5 Terms and Conditions

Display the applicable programme and website terms.

17.6 Privacy Policy

Explain the collection, use, storage and protection of customer information.

A separate Refund/Return Policy page is not required under the current website requirements. However, the no-cancellation, no-return and no-replacement condition must be clearly displayed before checkout.

18. WEBSITE DESIGN REQUIREMENTS

The website should have a premium, minimalist, Apple-inspired design.

The design must not copy Apple's logo or proprietary website layout.

Design requirements:

- Clean and modern appearance.
- Professional typography.
- Consistent spacing.
- Elegant product presentation.
- Smooth transitions.
- Simple navigation.
- Clear buttons and labels.
- Professional dashboard cards.
- Responsive tables.
- Consistent colours and typography.

The website must work properly on:

- Desktop
- Laptop
- Tablet
- Mobile

Plan A and Plan B should have consistent visual branding while maintaining separate programme dashboards.

19. TECHNICAL ARCHITECTURE

19.1 Backend

Technology:

Python + FastAPI

Backend responsibilities:

- Authentication
- Registration
- Referral code generation
- Referral validation
- Order processing
- Billing ID generation
- Commission calculation
- Wallet management
- Withdrawal processing
- Admin management
- Report generation
- Notification handling

19.2 Database

Database:

TiDB Cloud

Use a properly structured relational database.

Suggested database tables:

1. users
2. programmes
3. customer_profiles
4. referral_codes
5. referral_relationships
6. products
7. inventory
8. orders
9. order_items
10. billing_ids
11. commissions
12. wallets
13. wallet_transactions
14. withdrawals
15. wallet_freeze_history

16. account_status_history
17. admin_users
18. audit_logs
19. notifications
20. delivery_status

The developer may adjust the schema based on implementation requirements, but all business records and relationships must be preserved.

19.3 Separate Programme Data

Every relevant record must identify its programme.

Use programme identifiers to ensure that Plan A and Plan B records cannot be mixed.

For example:

- PLAN_A
- PLAN_B

Commission calculations must always validate the programme associated with the order, referral and Billing ID.

19.4 Security

Implement:

- Secure password hashing.
- Role-based access control.
- HTTPS.
- Secure database connections.
- Environment variables for secrets.
- Backend validation.
- SQL injection protection through safe database access.
- Authentication and authorization checks.
- Secure admin access.
- Audit logging.

19.5 Automatic Database Backups

Configure automatic database backups.

Ensure that backups can be used to restore the application data if required.

19.6 Development and Production Environments

Maintain separate environments:

Development:

- Testing
- Debugging
- Sample customers
- Sample orders

- Commission verification

Production:

- Real customer accounts
- Real business records
- Production database
- Production environment variables

Do not use development test data in production.

20. FINANCIAL AND TRANSACTION SAFETY

All financial calculations must use decimal-safe arithmetic.

Do not use floating-point arithmetic for financial balances.

Use database transactions when:

- Creating orders.
- Generating Billing IDs.
- Crediting commissions.
- Updating wallets.
- Processing withdrawals.

If a transaction fails, the database must not be left in a partially updated financial state.

20.1 Duplicate Commission Protection

The system must prevent duplicate commission payments.

Example:

If the same payment confirmation is received more than once:

- Do not create duplicate Billing IDs.
- Do not credit ₹500 again.
- Do not credit ₹50 again.
- Do not create duplicate wallet transactions.

Use unique references, database constraints and idempotency checks.

20.2 Permanent Audit Records

Maintain permanent audit records for:

- Commission credits.
- Wallet transactions.
- Withdrawals.
- Account blocking and deactivation.
- Wallet freezing and unfreezing.
- Admin actions.
- Referral relationships.

21. IMPORTANT IMPLEMENTATION RULES

The developer must follow these rules:

1. Do not combine Plan A and Plan B accounts.
2. Do not combine Plan A and Plan B wallets.
3. Do not mix commission calculations.
4. Do not allow referral code modification.
5. Do not allow admin to change referral relationships.
6. Do not credit ₹500 Direct Referral commission more than once per referred customer.
7. Credit ₹50 for every eligible Billing ID according to the confirmed Plan B rule.
8. Generate one unique Billing ID per Power Bank.
9. Do not generate Billing IDs for unsuccessful orders.
10. Do not generate duplicate commission transactions.
11. Do not allow partial withdrawals.
12. Enforce the first withdrawal minimum of ₹750.
13. Apply no minimum limit after the first successful withdrawal.
14. Allow automatic withdrawal only for eligible, active and unfrozen wallets.
15. Require admin review for withdrawal requests submitted by blocked distributors.
16. Do not permit direct withdrawals from frozen wallets.
17. Preserve all commission and wallet audit records.
18. Do not allow customers to cancel, return or replace products under the current rules.
19. Keep the current payment simulation separate from future real payment gateway integration.
20. Never invent additional commission recipients or generation-level payouts.

22. ITEMS REQUIRING FINAL BUSINESS CONFIRMATION

The following points should be resolved before production implementation.

1. Exact Generation Commission Allocation

The current confirmed rule is ₹50 for every eligible Billing ID to the applicable Plan B distributor. The original Plan B document describes five generations, but the newer instructions do not fully establish how five separate generation levels are assigned or whether any other person receives a share.

The developer must not assume additional recipients.

2. Inventory Updates

Admin cannot edit stock quantities under the current rule. However, the system must show In Stock or Out of Stock and prevent ordering when stock is unavailable.

The company must define how stock is replenished or updated.

3. Delivery Management

The system must notify distributors when an order is delivered. The method by which delivery status is updated must be finalized.

4. Deployment Architecture

The backend must use FastAPI and the database must use TiDB Cloud. Frontend and backend must not be deployed as separate services under the current instruction. The final hosting architecture must be selected before production deployment.

5. Registration Verification

The exact OTP and email verification requirements have not yet been finalized.

6. Real Payment Gateway

The first version uses simulated payment success when Place Order is clicked. Real payment gateway integration and payment methods will be decided later.

23. FINAL DEVELOPMENT OBJECTIVE

The final Plan B implementation must provide:

- Separate Plan B registration and login.
- Unique permanent referral codes.
- Referral code validation during checkout.
- Fixed product price of ₹3,999.
- One Billing ID per successfully purchased Power Bank.
- ₹500 Direct Referral commission once per referred customer.
- ₹50 Generation Commission for every eligible Billing ID.
- Automatic commission crediting.
- Independent distributor wallets.
- First withdrawal minimum of ₹750.
- No minimum withdrawal amount after the first successful withdrawal.
- Full wallet balance withdrawal only.
- Automatic withdrawals for eligible accounts.
- Blocked distributor withdrawal request management.
- Wallet freeze/unfreeze functionality.
- Distributor dashboard.
- Admin dashboard.
- Referral hierarchy inspection.
- Individual and consolidated Excel reports.
- Product and order management.
- Duplicate transaction protection.
- Permanent financial audit records.
- Responsive, premium website design.
- Secure backend and database architecture.

The implementation must follow the business rules in this document.

If a rule is unclear or contradictory, the developer must ask for clarification before implementing it. Do not silently invent business logic.

END OF PLAN B SOFTWARE DEVELOPMENT REQUIREMENTS

Arya Global Digital – Arya Generation Growth Programme

Prepared for: Arya Global Digital
Purpose: AI-assisted software development, implementation, testing and production deployment
Deployment: Render frontend + Render backend + TiDB Cloud database

26. FINAL CLARIFICATIONS AND IMPLEMENTATION INSTRUCTIONS

26.1 Combined Plan A and Plan B Website

The final website must support both programmes:

1. Plan A – Arya Business Partner Programme.
2. Plan B – Arya Generation Growth Programme.

Both programmes must operate within one Arya Global Digital website and one Admin management system.

However, their referral rules, commission calculations, wallets, cashback records, dashboards and reports must remain separate.

Important correction: The instruction on page 2 stating that only Plan A should be implemented and Plan B must not be implemented is no longer applicable. Both Plan A and Plan B must be implemented according to their respective sections in this document.

26.2 Plan A Commission and Cashback

- Direct referral commission: ₹750 gross per eligible Billing ID.
- Operational cost: ₹35.
- TDS: ₹15.
- Net commission: ₹700.
- Cashback contribution: ₹250 per eligible contribution.
- Cashback completion target: ₹4,000.
- Each completed cashback cycle requires 16 contributions.
- The dashboard must display 48 bulbs, with 3 bulbs activated for each ₹250 contribution.
- F1 and J1 cashback queues must operate independently using FIFO logic.
- Commission and cashback must have separate balances and transaction histories.

26.3 Plan B Commission

- Direct referral commission: ₹500 once per referred customer, after their first successful purchase.
- Generation commission: ₹50 for every eligible Billing ID.
- The ₹500 direct commission must not be credited again for future purchases by the same referred customer.
- The ₹50 generation commission must continue for every eligible Billing ID.
- Do not invent additional commission payments to multiple upstream distributors. Follow the confirmed direct-referral Billing ID rule unless the company provides a new written generation-wise distribution rule.

26.4 Important Business Rules

- Every successfully purchased Power Bank must receive its own unique Billing ID.

- A failed or pending payment must not generate a Billing ID or commission.
- Plan A and Plan B commission calculations must never be applied twice to the same Billing ID.
- Referral relationships must remain permanent after a successful order.
- Financial transactions must be processed on the backend and stored permanently.
- Duplicate orders and commission credits must be prevented.
- Admin must not manually modify wallet balances or referral relationships where prohibited by the programme rules.

26.5 Items Requiring Final Confirmation

Before implementing the related functionality in production, obtain written confirmation for:

1. Plan A: Whether multiple Billing IDs owned by the same customer can contribute to the same cashback cycle.
2. Plan A: The exact rules for assigning new Billing IDs to the independent F1 and J1 queues.
3. Plan B: Whether the ₹50 generation commission goes entirely to the direct referring distributor or is distributed among multiple generations. The current confirmed rule is ₹50 per eligible Billing ID to the applicable distributor.
4. Product stock: How the Admin will update product availability when stock quantity is not editable.
5. Payment: The actual payment gateway and payment methods for production.
6. Deployment: The final approved frontend and backend hosting arrangement.
7. Tax: The applicable GST, TDS and operational deductions must be verified by the company's accountant or tax adviser.

26.6 Final Developer Instruction

Read the complete document before starting development. Implement both programmes according to their separate business rules. Do not assume missing rules, change commission calculations, or combine financial records between the programmes.

Develop the application in small stages. Test registration, referrals, purchases, Billing ID generation, commissions, cashback, wallets, withdrawals, Admin operations and reports.

Do not enable real customer payments or commission withdrawals until the relevant business rules, financial calculations and security checks have been verified.

Final objective: One complete Arya Global Digital website supporting both Plan A and Plan B with separate programme logic and a common Admin management system.